

- ### SECTION-D
- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain Swinburne's test. Describe advantage and disadvantages of Swinburne test?
- Q.37 Draw and explain speed-torque characteristics of DC motors.
- Q.38 Explain in detail the different types of three-phase transformer connections.

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Subject:- Electrical Machines - I

SECTION-A

Q.1 Magnetic flux is measured in_____

a) Tesla b) Weber
c) Henry d) Ampere-turn

Q.2 Reluctance in a magnetic circuit corresponds to _____ in an electric circuit.

a) Current b) Resistance
c) Voltage d) Power

Q.3 The product of current and flux linkage gives-

a) Power b) Energy
c) Torque d) Inductance

Q.4 The basic principle of DC machines is-

a) Static induction
b) Electromagnetic induction
c) Thermal effect
d) Photoelectric effect

(1) 180941/170941/120941
/030941

- Q.5 The purpose of brushes is to-
- Carry current to armature
 - Clean commutator
 - Produce torque
 - Control speed
- Q.6 ADC generator converts-
- Electrical to mechanical energy
 - Mechanical to electrical energy
 - AC to DC
 - DC to AC
- Q.7 The material used for brushes is-
- Copper
 - Iron
 - Graphite
 - Aluminum
- Q.8 In an ideal transformer, losses are-
- Maximum
 - Minimum
 - Zero
 - Infinite
- Q.9 In delta-delta connection, the phase shift between primary and secondary is-
- 0°
 - 30°
 - 60°
 - 90°
- Q.10 The open-delta connection can supply-
- Full load
 - 57.7% of full load
 - 75% of full load
 - 100% of full load

(2) 180941/170941/120941
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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define hysteresis loss.
- Q.12 What is the basic principle of transformer
- Q.13 State Lenz's law..
- Q.14 Write the formula for reluctance.
- Q.15 Define torque in DC motors.
- Q.16 What is the use of carbon brushes.
- Q.17 Define short-circuit test.
- Q.18 Define voltage regulation.
- Q.19 Mention one advantage of three-phase transformer.
- Q.20 Name the cooling methods used in three-phase transformer.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the principle of electromechanical energy conversion.
- Q.22 Describe hysteresis and eddy current losses in detail.
- Q.23 Explain how electromagnetic torque is produced.
- Q.24 Write the methods of speed control of dc shunt motor. Explain any one.
- Q.25 Differentiate step-up and step-down transformer.
- Q.26 Can a transformer work on d c supply? Discuss.
- Q.27 Draw a phasor diagram of 1-phase transformer for inductive load.

(3) 180941/170941/120941
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